

Interval Belief Propagation for Credal Networks with Multi-Valued Variables

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Abstract

Interval Belief Propagation (IBP) is an iterative message-passing algorithm for approximate marginal inference in credal networks. Unlike the 2U and L2U algorithms which are restricted to binary variables, IBP operates natively on multi-valued variables by propagating interval-vector messages. At each iteration, variable nodes tighten their interval bounds by intersecting incoming messages, while factor nodes compute new bounds by optimizing over the extreme points of local credal sets and the corner points of incoming interval messages. The algorithm is inspired by the ApproxLP framework of Antonucci et al. (2015) and the modified belief propagation for Logical Credal Networks (Qian et al., 2021).

1 Preliminaries

1.1 Factor Graph

Given a credal network $\langle G, \mathcal{K} \rangle$, we construct a **factor graph**: a bipartite graph with variable nodes and factor nodes. Each variable X_i in the credal network corresponds to a variable node with cardinality $|D_i|$. Each conditional credal set $K(X_i \mid \text{Pa}(X_i))$ defines a factor node f_i whose scope is $\{X_i\} \cup \text{Pa}(X_i)$. The factor stores the extreme points of $K(X_i \mid \pi)$ for all parent configurations π .

1.2 Interval Messages

Definition 1 (Interval Message). *For a variable X with k states, an **interval message** is a pair of vectors $(\mathbf{l}, \mathbf{u}) \in \mathbb{R}^k \times \mathbb{R}^k$ where $0 \leq l_x \leq u_x \leq 1$ for each state $x \in D_X$. The interval $[l_x, u_x]$ bounds the probability $P(X = x)$ from the message sender’s perspective.*

Messages are of two types:

- $\mu_{v \rightarrow f} = (\mathbf{l}_{v \rightarrow f}, \mathbf{u}_{v \rightarrow f})$: message from variable v to factor f .
- $\mu_{f \rightarrow v} = (\mathbf{l}_{f \rightarrow v}, \mathbf{u}_{f \rightarrow v})$: message from factor f to variable v .

All messages are initialized to the vacuous interval $([0, \dots, 0], [1, \dots, 1])$.

1.3 Evidence

Evidence is a (possibly empty) set of observed variable assignments

$$\mathbf{Y} = \mathbf{y} = \{Y_1 = y_1, Y_2 = y_2, \dots, Y_m = y_m\}.$$

For each evidence variable $Y_j = y_j$, the outgoing variable-to-factor message is clamped to the one-hot indicator $\delta_{y_j} \in \{0, 1\}^{|D_{Y_j}|}$ (entry y_j equals 1, all others equal 0):

$$\mu_{Y_j \rightarrow f} \leftarrow (\delta_{y_j}, \delta_{y_j}) \quad \text{for every factor } f \in \mathcal{N}(Y_j).$$

The interval collapses to a single point, pinning Y_j to y_j . Two consequences for the main loop:

- **Clamped messages are held fixed.** During the variable-to-factor update phase, evidence variables are *skipped* — their outgoing messages are never recomputed.
- **Factor-to-variable updates use the clamped input.** When a factor f computes an outgoing message, any incoming message from an evidence variable Y_j has $\mathbf{l}_{Y_j \rightarrow f} = \mathbf{u}_{Y_j \rightarrow f} = \delta_{y_j}$. The corner-point enumeration of Section 4.2 then contributes no degrees of freedom for Y_j : both corners select the same vector $\mathbf{q}_{Y_j} = \delta_{y_j}$, and the marginal sum in Equations 4–5 is restricted to parent configurations consistent with $Y_j = y_j$.

After convergence, the extracted marginal bounds at every non-evidence variable X_i are bounds on the *conditional* marginal $P(X_i \mid \mathbf{Y} = \mathbf{y})$.

For compound variables that bundle multiple atoms, if an atom is itself an evidence variable, the singleton-marginal extraction LP (described in the companion CCTE document) skips that atom, since its bound is trivially $[\mathbb{K}[y_j=1], \mathbb{K}[y_j=1]]$.

2 Algorithm

Algorithm 1 Interval Belief Propagation (IBP)

Require: Factor graph with extreme points, evidence $\mathbf{Y} = \mathbf{y}$, max iterations T , threshold τ

Ensure: Bounds $[\underline{P}(X_i=x), \overline{P}(X_i=x)]$ for **all** variables X_i and states x

```

1: Initialize  $\mu_{v \rightarrow f} \leftarrow (\mathbf{0}, \mathbf{1})$  and  $\mu_{f \rightarrow v} \leftarrow (\mathbf{0}, \mathbf{1})$  for all edges
                                                                    ▷ EVIDENCE CLAMPING
2: for each evidence variable  $Y_j = y_j \in \mathbf{Y}$  do
3:   Set  $\mu_{Y_j \rightarrow f} \leftarrow (\delta_{y_j}, \delta_{y_j})$  for all factors  $f \in \mathcal{N}(Y_j)$     ▷  $\delta_{y_j}$  is the one-hot indicator at  $y_j$ 
4: end for

5: for  $t = 1, \dots, T$  do
6:    $\delta \leftarrow 0$ 
                                                                    ▷ VARIABLE-TO-FACTOR UPDATES
7:   for each variable  $v$ , each factor  $f \in \mathcal{N}(v)$  do
8:     if  $v \in \mathbf{Y}$  then continue    ▷ evidence messages are held fixed
9:   end if
10:  for each state  $x$  do
11:     $l_{v \rightarrow f}(x) \leftarrow \max_{f' \in \mathcal{N}(v) \setminus \{f\}} l_{f' \rightarrow v}(x)$ 
12:     $u_{v \rightarrow f}(x) \leftarrow \min_{f' \in \mathcal{N}(v) \setminus \{f\}} u_{f' \rightarrow v}(x)$ 
13:  end for
14:   $\mathbf{u}_{v \rightarrow f} \leftarrow \max(l_{v \rightarrow f}, \mathbf{u}_{v \rightarrow f})$     ▷ Ensure  $l \leq u$ 
15:  Update  $\delta$ 
16: end for
                                                                    ▷ FACTOR-TO-VARIABLE UPDATES
17: for each factor  $f_i$ , each target variable  $v \in \text{scope}(f_i)$  do
18:    $\mathbf{l}_{\text{new}} \leftarrow \mathbf{1}, \quad \mathbf{u}_{\text{new}} \leftarrow \mathbf{0}$ 
19:   for each vertex combination vc do
20:     for each corner-point combination cc do
21:       Compute unnormalized marginal  $\mathbf{m}(v)$  using vc and cc
22:        $\mathbf{p} \leftarrow \mathbf{m} / \sum_x m(x)$ 
23:        $\mathbf{l}_{\text{new}} \leftarrow \min(\mathbf{l}_{\text{new}}, \mathbf{p})$ 
24:        $\mathbf{u}_{\text{new}} \leftarrow \max(\mathbf{u}_{\text{new}}, \mathbf{p})$ 
25:     end for
26:   end for
27:    $\mu_{f \rightarrow v} \leftarrow (\mathbf{l}_{\text{new}}, \mathbf{u}_{\text{new}})$ 
28:   Update  $\delta$ 
29: end for
30: if  $\delta < \tau$  then break
31: end if
32: end for
                                                                    ▷ EXTRACT MARGINAL BOUNDS (ALL VARIABLES, GIVEN EVIDENCE)
33: for each variable  $X_i$  in the factor graph do
34:   for each state  $x \in D_{X_i}$  do
35:      $\underline{P}(X_i=x \mid \mathbf{Y}=\mathbf{y}) \leftarrow \max_{f \in \mathcal{N}(X_i)} l_{f \rightarrow X_i}(x)$ 
36:      $\overline{P}(X_i=x \mid \mathbf{Y}=\mathbf{y}) \leftarrow \min_{f \in \mathcal{N}(X_i)} u_{f \rightarrow X_i}(x)$ 
37:   end for
38: end for
39: return  $[\underline{P}(X_i=x \mid \mathbf{Y}=\mathbf{y}), \overline{P}(X_i=x \mid \mathbf{Y}=\mathbf{y})]$  for every variable  $X_i$  and state  $x$ 

```

3 Compact Description (NeurIPS Format)

Setup. Given a credal network $\langle G, \mathcal{K} \rangle$ compiled to a factor graph whose factor nodes $\{f_i\}$ store the finite sets of extreme points $\text{ext } K(X_i \mid \pi)$ (one per parent configuration π of the associated variable X_i), plus evidence $\mathbf{Y} = \mathbf{y}$, *interval belief propagation* (IBP) computes bounds $[\underline{P}(X_i=x \mid \mathbf{y}), \overline{P}(X_i=x \mid \mathbf{y})]$ for every non-evidence variable X_i and every state $x \in D_{X_i}$.

Messages. Each directed edge of the factor graph carries an interval-vector message $\mu_{v \rightarrow f} = (\mathbf{l}_{v \rightarrow f}, \mathbf{u}_{v \rightarrow f}) \in [0, 1]^{|D_v|} \times [0, 1]^{|D_v|}$ (and symmetrically $\mu_{f \rightarrow v}$) with $\mathbf{l} \leq \mathbf{u}$ componentwise. All messages are initialized to the vacuous interval $(\mathbf{0}, \mathbf{1})$. For every evidence assignment $Y_j = y_j$, the outgoing variable-to-factor messages from Y_j are clamped to the one-hot indicator $\mu_{Y_j \rightarrow f} \leftarrow (\delta_{y_j}, \delta_{y_j})$ and held fixed throughout.

Updates. At every iteration, two sweeps are performed. *Variable-to-factor* messages (skipped for evidence variables) tighten by intersecting the other incoming bounds:

$$\mathbf{l}_{v \rightarrow f} = \max_{f' \in \mathcal{N}(v) \setminus \{f\}} \mathbf{l}_{f' \rightarrow v}, \quad \mathbf{u}_{v \rightarrow f} = \min_{f' \in \mathcal{N}(v) \setminus \{f\}} \mathbf{u}_{f' \rightarrow v}, \quad v \notin \mathbf{Y} \quad (1)$$

(componentwise, over all neighbor factors other than f). *Factor-to-variable* messages extremize the target's normalized marginal over two discrete choice sets:

$$\mathbf{l}_{f_i \rightarrow v}(x) = \min_{(\mathbf{vc}, \mathbf{cc})} \tilde{p}_{\mathbf{vc}, \mathbf{cc}}(x), \quad \mathbf{u}_{f_i \rightarrow v}(x) = \max_{(\mathbf{vc}, \mathbf{cc})} \tilde{p}_{\mathbf{vc}, \mathbf{cc}}(x), \quad (2)$$

where a *vertex combination* $\mathbf{vc}$ picks one extreme point $P_{\mathbf{vc}}(X_i \mid \pi) \in \text{ext } K(X_i \mid \pi)$ for every parent configuration π , a *corner-point combination* $\mathbf{cc}$ picks $\mathbf{q}_w \in \{\mathbf{l}_{w \rightarrow f_i}, \mathbf{u}_{w \rightarrow f_i}\}$ for each other-variable $w \in \text{scope}(f_i) \setminus \{v\}$, and $\tilde{p}_{\mathbf{vc}, \mathbf{cc}}(x) = m_{\mathbf{vc}, \mathbf{cc}}(x) / \sum_{x'} m_{\mathbf{vc}, \mathbf{cc}}(x')$ with $m_{\mathbf{vc}, \mathbf{cc}}$ the unnormalized marginal given by Eq. 4 (if $v = X_i$, the child) or Eq. 5 (if $v \in \text{Pa}(X_i)$, a parent).

Convergence and extraction. The two sweeps are repeated until the maximum componentwise change across all messages falls below a tolerance τ , or a cap of T iterations is reached. The final conditional marginal bounds for every variable are obtained by intersecting all factor-to-variable messages incident on it:

$$\underline{P}(X_i=x \mid \mathbf{y}) = \max_{f \in \mathcal{N}(X_i)} \mathbf{l}_{f \rightarrow X_i}(x), \quad \overline{P}(X_i=x \mid \mathbf{y}) = \min_{f \in \mathcal{N}(X_i)} \mathbf{u}_{f \rightarrow X_i}(x). \quad (3)$$

Algorithm 2 Interval Belief Propagation — Compact Form

Require: Factor graph with extreme points, evidence $\mathbf{Y} = \mathbf{y}$, max iterations T , tolerance τ

Ensure: $[\underline{P}(X_i=x \mid \mathbf{y}), \overline{P}(X_i=x \mid \mathbf{y})]$ for all $X_i \notin \mathbf{Y}$, $x \in D_{X_i}$

- 1: Initialize $\mu_{v \rightarrow f}, \mu_{f \rightarrow v} \leftarrow (\mathbf{0}, \mathbf{1})$ for every edge; clamp $\mu_{Y_j \rightarrow f} \leftarrow (\delta_{y_j}, \delta_{y_j})$ for each $Y_j \in \mathbf{Y}$
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: Update $\mu_{v \rightarrow f}$ via Eq. 1 for every edge (v, f) with $v \notin \mathbf{Y}$
 - 4: Update $\mu_{f \rightarrow v}$ via Eq. 2 for every edge (f, v)
 - 5: **if** max componentwise change in any message $< \tau$ **then break**
 - 6: **end if**
 - 7: **end for**
 - 8: **return** bounds via Eq. 3 for every $X_i \notin \mathbf{Y}$
-

4 Factor-to-Variable Message Computation

The factor-to-variable message from factor f_i (associated with node X_i , parents $\text{Pa}(X_i)$) to target variable $v \in \text{scope}(f_i)$ is computed by scanning over all *vertex combinations* and *corner-point combinations*, as defined below.

Let $\mathcal{O} = \text{scope}(f_i) \setminus \{v\}$ be the other variables. For each $w \in \mathcal{O}$, let $(\mathbf{l}_w, \mathbf{u}_w) = \mu_{w \rightarrow f_i}$ be the incoming message.

4.1 Vertex Combinations (vc)

Factor f_i stores a conditional credal set $K(X_i \mid \pi)$ for each parent configuration $\pi \in \Omega_{\text{Pa}(X_i)}$. Each credal set has a finite number of extreme points (vertices):

$$\text{ext } K(X_i \mid \pi) = \{P_1(X_i \mid \pi), P_2(X_i \mid \pi), \dots, P_{n_\pi}(X_i \mid \pi)\}.$$

Definition 2 (Vertex Combination). A **vertex combination** $\mathbf{vc} = (j_1, j_2, \dots, j_m)$ selects one vertex per parent configuration: vertex j_k from configuration π_k . It specifies a complete conditional distribution $P_{\mathbf{vc}}(X_i \mid \pi_k)$ for every parent configuration.

The total number of vertex combinations is $\prod_{k=1}^m n_{\pi_k}$.

Example. If factor f_A has parents $\{B, E\}$ with 4 parent configurations $(B=0, E=0)$, $(B=1, E=0)$, $(B=0, E=1)$, $(B=1, E=1)$, and each has 2 vertices, then there are $2^4 = 16$ vertex combinations. The combination $\mathbf{vc} = (1, 1, 1, 1)$ selects vertex 1 from every parent configuration, while $\mathbf{vc} = (2, 1, 2, 1)$ selects vertex 2 from $(B=0, E=0)$ and $(B=0, E=1)$, and vertex 1 from the other two.

4.2 Corner-Point Combinations (cc)

For each “other” variable $w \in \mathcal{O}$, the incoming interval message provides a lower bound vector $\mathbf{l}_w$ and an upper bound vector $\mathbf{u}_w$. To compute the factor-to-variable message, we must evaluate the marginal at the extreme points of these intervals.

Definition 3 (Corner-Point Combination). A **corner-point combination** $\mathbf{cc} = (c_1, c_2, \dots, c_{|\mathcal{O}|})$ where each $c_j \in \{0, 1\}$ selects either the lower ($c_j = 0$) or upper ($c_j = 1$) bound vector for each other variable w_j :

$$\mathbf{q}_{w_j} = \begin{cases} \mathbf{l}_{w_j} & \text{if } c_j = 0 \\ \mathbf{u}_{w_j} & \text{if } c_j = 1 \end{cases}$$

The total number of corner-point combinations is $2^{|\mathcal{O}|}$.

Example. When computing $f_A \rightarrow B$ (target is B), the other variables are $\mathcal{O} = \{A, E\}$. If the incoming messages are $\mu_{A \rightarrow f_A} = ([0.7, 0.0], [1.0, 0.3])$ and $\mu_{E \rightarrow f_A} = ([0.9, 0.05], [0.95, 0.1])$, then there are $2^2 = 4$ corner-point combinations selecting $\mathbf{q}_A \in \{[0.7, 0.0], [1.0, 0.3]\}$ and $\mathbf{q}_E \in \{[0.9, 0.05], [0.95, 0.1]\}$.

4.3 Marginal Computation

For a given $(\mathbf{vc}, \mathbf{cc})$ pair, the unnormalized marginal over v is:

Case 1: $v = X_i$ (target is the child node):

$$m(v=x) = \sum_{\pi} P_{\mathbf{vc}}(X_i=x \mid \pi) \cdot \prod_{w \in \text{Pa}(X_i) \cap \mathcal{O}} \mathbf{q}_w(\pi_w), \quad (4)$$

where $\mathbf{q}_w$ is the bound vector for w selected by $\mathbf{cc}$, and π_w is the value of w in parent configuration π .

Case 2: $v \in \text{Pa}(X_i)$ (target is a parent):

$$m(v=x) = \sum_{\pi: \pi_v=x} \left(\sum_{x_i} P_{\mathbf{vc}}(X_i=x_i \mid \pi) \cdot \mathbf{q}_{X_i}(x_i) \right) \cdot \prod_{w \in \text{Pa} \setminus \{v\}} \mathbf{q}_w(\pi_w), \quad (5)$$

where $\mathbf{q}_{X_i}$ is the bound vector for X_i if $X_i \in \mathcal{O}$.

The marginal is normalized: $p(x) = m(x) / \sum_{x'} m(x')$.

The factor-to-variable message then takes the componentwise minimum and maximum over all $(\mathbf{vc}, \mathbf{cc})$ pairs:

$$l_{f \rightarrow v}(x) = \min_{\mathbf{vc}, \mathbf{cc}} p(x),$$

$$u_{f \rightarrow v}(x) = \max_{\mathbf{vc}, \mathbf{cc}} p(x).$$

5 Detailed Running Example: Alarm Network

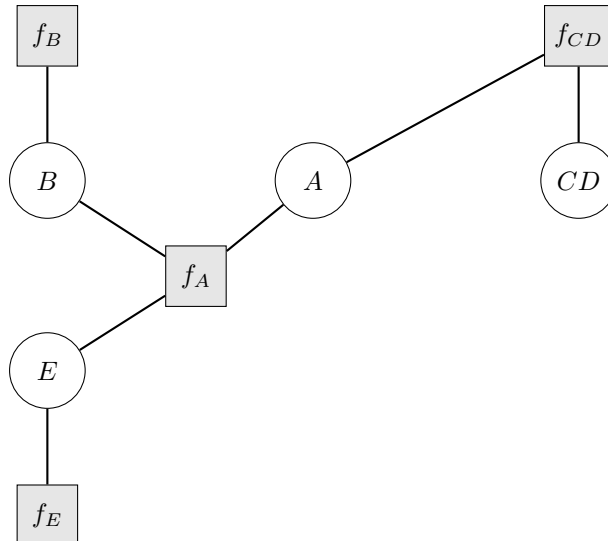
We trace the IBP algorithm on the alarm network from file `alarm.lcn`:

```
s1: 0.1 <= P(B) <= 0.2
s2: 0.05 <= P(E) <= 0.1
s3: 0.6 <= P(A | (B or E)) <= 0.9
s4: 0.5 <= P(not(C xor D) | A) <= 0.7 ; True
s5: 0.05 <= P(A) <= 0.1
```

After the chain graph factorization and LRS vertex enumeration, the credal network has four nodes: B (binary), E (binary), A (binary), and CD (4-valued, representing the compound variable for C and D). The DAG structure is $B \rightarrow A$, $E \rightarrow A$, $A \rightarrow CD$.

5.1 Factor Graph

The factor graph is a bipartite graph with 4 variable nodes and 4 factor nodes:



Each factor's scope and adjacency:

Factor	Node	Parents	Scope
f_B	B	—	$\{B\}$
f_E	E	—	$\{E\}$
f_A	A	B, E	$\{A, B, E\}$
f_{CD}	CD	A	$\{CD, A\}$

Variable adjacency: $B \leftrightarrow \{f_B, f_A\}$, $E \leftrightarrow \{f_E, f_A\}$, $A \leftrightarrow \{f_A, f_{CD}\}$, $CD \leftrightarrow \{f_{CD}\}$.

5.2 Credal Factors (Extreme Points)

Factor f_B : $K(B)$, no parents.

Vertex	$P(B=0)$	$P(B=1)$
v_1	0.80	0.20
v_2	0.90	0.10

Factor f_E : $K(E)$, no parents.

Vertex	$P(E=0)$	$P(E=1)$
v_1	0.90	0.10
v_2	0.95	0.05

Factor f_A : $K(A \mid B, E)$, **parents** $\{B, E\}$. Four parent configurations, each with 2 vertices ($2^4 = 16$ vertex combinations):

Parent config	Vertex	$P(A=0 \mid \cdot)$	$P(A=1 \mid \cdot)$
$B=0, E=0$	v_1	0.9556	0.0444
$B=0, E=0$	v_2	1.0000	0.0000
$B=1, E=0$	v_1	0.0000	1.0000
$B=1, E=0$	v_2	1.0000	0.0000
$B=0, E=1$	v_1	0.0000	1.0000
$B=0, E=1$	v_2	1.0000	0.0000
$B=1, E=1$	v_1	0.0000	1.0000
$B=1, E=1$	v_2	1.0000	0.0000

Factor f_{CD} : $K(CD \mid A)$, **parent** $\{A\}$. Two parent configurations: $A=0$ (4 vertices) and $A=1$ (11 vertices). For brevity, we show a subset:

Config	Vtx	$P(CD=0)$	$P(CD=1)$	$P(CD=2)$	$P(CD=3)$
$A=0$	v_1	0.00	0.00	0.00	1.00
$A=0$	v_2	0.00	0.00	1.00	0.00
$A=0$	v_3	0.00	1.00	0.00	0.00
$A=0$	v_4	1.00	0.00	0.00	0.00
$A=1$	v_1	0.00	0.00	0.30	0.70
$A=1$	v_2	0.00	0.30	0.00	0.70
$\dots$ (9 more)					

There are $4 \times 11 = 44$ vertex combinations for f_{CD} .

5.3 Query: $P(B=1)$, No Evidence

5.3.1 Iteration 1: Factor-to-Variable Messages

Message $f_B \rightarrow B$. Factor f_B has scope $\{B\}$ with no parents and no other variables ($\mathcal{O} = \emptyset$).

- **Vertex combinations:** 2 (one per vertex of $K(B)$, since there is only one parent configuration—the empty one).
- **Corner-point combinations:** $2^0 = 1$ (no other variables).

Computation for each **vc**:

$$\begin{aligned} \mathbf{vc} = (1) : \quad m = v_1 &= [0.80, 0.20], \quad p = m/1.0 = [0.80, 0.20]. \\ \mathbf{vc} = (2) : \quad m = v_2 &= [0.90, 0.10], \quad p = m/1.0 = [0.90, 0.10]. \end{aligned}$$

Result:

$$\mu_{f_B \rightarrow B} = \left(\underbrace{[0.80, 0.10]}_{\mathbf{l}=\min}, \underbrace{[0.90, 0.20]}_{\mathbf{u}=\max} \right).$$

Message $f_E \rightarrow E$. Analogous computation:

$$\mu_{f_E \rightarrow E} = \left([0.90, 0.05], [0.95, 0.10] \right).$$

Message $f_A \rightarrow A$. Factor f_A has scope $\{A, B, E\}$, target is A , so $\mathcal{O} = \{B, E\}$. This is Case 1 (target is the child node). Using Equation 4:

$$m(A=x) = \sum_{\pi \in \{0,1\}^2} P_{\mathbf{vc}}(A=x \mid B=\pi_B, E=\pi_E) \cdot \mathbf{q}_B(\pi_B) \cdot \mathbf{q}_E(\pi_E).$$

In iteration 1, the incoming messages are vacuous: $\mu_{B \rightarrow f_A} = ([0, 0], [1, 1])$, $\mu_{E \rightarrow f_A} = ([0, 0], [1, 1])$.

- **Vertex combinations:** $2^4 = 16$ (2 vertices per each of 4 parent configs).
- **Corner-point combinations:** $2^2 = 4$ (lower or upper for B and E).

When both incoming messages are vacuous, many $(\mathbf{vc}, \mathbf{cc})$ pairs produce $m = [0, 0]$ (zero total) because $\mathbf{q}_B = [0, 0]$ or $\mathbf{q}_E = [0, 0]$ zeroes out the product. Only $\mathbf{cc} = (1, 1)$ (both upper bounds $= [1, 1]$) produces nonzero marginals.

Detailed example: $\mathbf{vc} = (1, 1, 1, 1)$ (vertex 1 for all configs), $\mathbf{cc} = (1, 1)$ ($\mathbf{q}_B = [1, 1]$, $\mathbf{q}_E = [1, 1]$):

$$\begin{aligned} m(A=0) &= 0.9556 \cdot 1 \cdot 1 + 0.0000 \cdot 1 \cdot 1 + 0.0000 \cdot 1 \cdot 1 + 0.0000 \cdot 1 \cdot 1 = 0.9556, \\ m(A=1) &= 0.0444 \cdot 1 \cdot 1 + 1.0000 \cdot 1 \cdot 1 + 1.0000 \cdot 1 \cdot 1 + 1.0000 \cdot 1 \cdot 1 = 3.0444, \\ p &= [0.9556, 3.0444]/4.0 = [0.2389, 0.7611]. \end{aligned}$$

After scanning all 16 vertex combinations (with $\mathbf{cc} = (1, 1)$), the resulting message is wide because vacuous inputs provide no information.

Message $f_A \rightarrow B$. Target is B (a parent), $\mathcal{O} = \{A, E\}$. This is Case 2. With vacuous incoming messages, the message to B is similarly wide.

Messages from f_{CD} . Factor f_{CD} has scope $\{CD, A\}$ with 44 vertex combinations and $2^1 = 2$ corner-point combinations per target variable. In the first iteration with vacuous incoming messages, these produce wide bounds.

5.3.2 Iteration 1: Variable-to-Factor Messages

After the factor-to-variable messages are computed, the variable-to-factor messages are updated by *tightening*: each variable sends to a factor the intersection of messages from all *other* factors.

Message $B \rightarrow f_A$. Variable B has neighbors $\{f_B, f_A\}$. The message to f_A excludes f_A itself:

$$\mu_{B \rightarrow f_A} = \mu_{f_B \rightarrow B} = ([0.80, 0.10], [0.90, 0.20]).$$

This is a significant tightening from the vacuous $([0, 0], [1, 1])$: factor f_B directly constrains B 's marginal, and this information is now passed to f_A .

Message $E \rightarrow f_A$. Similarly, E has neighbors $\{f_E, f_A\}$:

$$\mu_{E \rightarrow f_A} = \mu_{f_E \rightarrow E} = ([0.90, 0.05], [0.95, 0.10]).$$

Message $A \rightarrow f_{CD}$. Variable A has neighbors $\{f_A, f_{CD}\}$:

$$\mu_{A \rightarrow f_{CD}} = \mu_{f_A \rightarrow A}.$$

5.3.3 Iteration 2: Recomputing $f_A \rightarrow A$ with Informative Messages

Now f_A receives informative messages from B and E . The computation of $f_A \rightarrow A$ uses:

$$\begin{aligned} \mathbf{q}_B &\in \{[0.80, 0.10], [0.90, 0.20]\}, \\ \mathbf{q}_E &\in \{[0.90, 0.05], [0.95, 0.10]\}. \end{aligned}$$

Detailed example: $\mathbf{vc} = (1, 1, 1, 1)$, $\mathbf{cc}$: $\mathbf{q}_B = [0.80, 0.10]$, $\mathbf{q}_E = [0.90, 0.05]$:

$$\begin{aligned} m(A=0) &= 0.9556 \cdot 0.80 \cdot 0.90 + 0.0000 \cdot 0.10 \cdot 0.90 \\ &\quad + 0.0000 \cdot 0.80 \cdot 0.05 + 0.0000 \cdot 0.10 \cdot 0.05 = 0.6880, \\ m(A=1) &= 0.0444 \cdot 0.80 \cdot 0.90 + 1.0000 \cdot 0.10 \cdot 0.90 \\ &\quad + 1.0000 \cdot 0.80 \cdot 0.05 + 1.0000 \cdot 0.10 \cdot 0.05 = 0.1670, \\ p &= [0.6880, 0.1670] / 0.8550 = [0.8047, 0.1953]. \end{aligned}$$

Another example: $\mathbf{vc} = (1, 1, 1, 1)$, $\mathbf{cc}$: $\mathbf{q}_B = [0.90, 0.20]$, $\mathbf{q}_E = [0.95, 0.10]$:

$$\begin{aligned} m(A=0) &= 0.9556 \cdot 0.90 \cdot 0.95 + 0.0 \cdot 0.20 \cdot 0.95 \\ &\quad + 0.0 \cdot 0.90 \cdot 0.10 + 0.0 \cdot 0.20 \cdot 0.10 = 0.8170, \\ m(A=1) &= 0.0444 \cdot 0.90 \cdot 0.95 + 1.0 \cdot 0.20 \cdot 0.95 \\ &\quad + 1.0 \cdot 0.90 \cdot 0.10 + 1.0 \cdot 0.20 \cdot 0.10 = 0.3380, \\ p &= [0.8170, 0.3380] / 1.1550 = [0.7074, 0.2926]. \end{aligned}$$

As the incoming messages become more informative (tighter intervals), the factor explores a more constrained space and the message to A tightens accordingly.

5.3.4 Final Marginal Extraction

After convergence, the marginal bounds for B are:

$$\underline{P}(B=x) = \max_{f \in \{f_B, f_A\}} l_{f \rightarrow B}(x), \quad \overline{P}(B=x) = \min_{f \in \{f_B, f_A\}} u_{f \rightarrow B}(x).$$

Factor f_B provides the tightest bounds for B since it directly encodes $K(B)$:

$$P(B=1) \in [0.10, 0.20].$$

5.4 Query with Evidence: $P(B=1 \mid E=1)$

We now rerun IBP with evidence $E = 1$. According to Section 1.3, the outgoing messages from E are clamped to the one-hot indicator for state 1:

$$\mu_{E \rightarrow f_E} = \mu_{E \rightarrow f_A} = ([0, 1], [0, 1]),$$

and E is skipped in every subsequent variable-to-factor update.

5.4.1 Iteration 1: $f_B \rightarrow B$ and $f_E \rightarrow E$.

f_B is not adjacent to E , so its outgoing message is identical to the no-evidence case:

$$\mu_{f_B \rightarrow B} = ([0.80, 0.10], [0.90, 0.20]).$$

$f_E \rightarrow E$ is also unchanged by the clamping of $\mu_{E \rightarrow f_E}$ (the factor receives evidence on its target; the outgoing message still reflects $K(E)$). However, this message will be intersected with the clamped $\mu_{E \rightarrow f_E}$ only at the extraction step, so in practice $P(E)$ is pinned to $[0, 1]$ on state 1.

5.4.2 Iteration 1: $f_A \rightarrow A$ with E clamped.

Target is A (Case 1), $\mathcal{O} = \{B, E\}$. In iteration 1, B still sends the vacuous message $\mu_{B \rightarrow f_A} = ([0, 0], [1, 1])$, but $\mu_{E \rightarrow f_A} = ([0, 1], [0, 1])$. The corner-point enumeration over $\mathcal{O}$ now collapses along the E axis since both corners select $\mathbf{q}_E = [0, 1]$, leaving only the two B corners free. Using Equation 4:

$$m(A=x) = \sum_{\pi_B, \pi_E \in \{0,1\}} P_{\mathbf{vc}}(A=x \mid B=\pi_B, E=\pi_E) \cdot \mathbf{q}_B(\pi_B) \cdot \underbrace{\mathbf{q}_E(\pi_E)}_{0 \text{ if } \pi_E=0}.$$

Only the $\pi_E = 1$ terms survive. With $\mathbf{vc} = (1, 1, 1, 1)$ (vertex 1 for every parent configuration) and $\mathbf{q}_B = [1, 1]$ (the upper corner, since B 's message is vacuous):

$$\begin{aligned} m(A=0) &= \underbrace{0.9556 \cdot 1 \cdot 0}_{B=0, E=0} + \underbrace{0.0000 \cdot 1 \cdot 0}_{B=1, E=0} + \underbrace{0.0000 \cdot 1 \cdot 1}_{B=0, E=1} + \underbrace{0.0000 \cdot 1 \cdot 1}_{B=1, E=1} = 0, \\ m(A=1) &= \underbrace{0.0444 \cdot 1 \cdot 0}_{B=0, E=0} + \underbrace{1.0000 \cdot 1 \cdot 0}_{B=1, E=0} + \underbrace{1.0000 \cdot 1 \cdot 1}_{B=0, E=1} + \underbrace{1.0000 \cdot 1 \cdot 1}_{B=1, E=1} = 2.0, \\ p &= [0, 2.0]/2.0 = [0, 1]. \end{aligned}$$

Under $\mathbf{vc} = (2, 2, 1, 1)$ (vertex 2 for $B=0, E=0$ and $B=1, E=0$, vertex 1 for $B=0, E=1$ and $B=1, E=1$), the same calculation still gives $p = [0, 1]$ because the $\pi_E = 0$ terms are zeroed out regardless of the vertex choice there. Clamping $E=1$ drives the marginal of A toward $A=1$: whenever $E=1$ forces the child, all surviving vertices in these configurations have $P(A=1 \mid \cdot) = 1.0$ for vertex 1 and $P(A=0 \mid \cdot) = 1.0$ for vertex 2. Combining across vertex combinations yields $\mu_{f_A \rightarrow A} = ([0, 0], [1, 1])$ in iteration 1 — still wide, because B is vacuous.

5.4.3 Iteration 1: $f_A \rightarrow B$ with E clamped.

Target is B (Case 2), $\mathcal{O} = \{A, E\}$. Again $\mathbf{q}_E = [0, 1]$ in both corners, and only parent configurations with $\pi_E = 1$ contribute. Using Equation 5 with $\mathbf{q}_A = [0, 0]$ or $[1, 1]$ (vacuous):

$$m(B=x) = \sum_{\pi_E=1} \left(\sum_{x_i} P_{\mathbf{vc}}(A=x_i \mid B=x, E=1) \cdot \mathbf{q}_A(x_i) \right) \cdot \mathbf{q}_E(1).$$

The inner sum over x_i is exactly $\sum_{x_i} \mathbf{q}_A(x_i) \cdot 1$ (since vertex 1 has $P(A=1 \mid \cdot) = 1$ and vertex 2 has $P(A=0 \mid \cdot) = 1$ for the two $E=1$ rows). The upper corner $\mathbf{q}_A = [1, 1]$ gives $m(B=0) = m(B=1) = 1$, i.e. $p = [0.5, 0.5]$; the lower corner $\mathbf{q}_A = [0, 0]$ zeros out the marginal. Combined with other vertex combinations, $\mu_{f_A \rightarrow B}$ in iteration 1 remains wide.

5.4.4 Iteration 2: $B \rightarrow f_A$ and recomputation.

By Section 1.3, E is skipped in the variable-to-factor phase; only B (and A, CD) are updated. B 's message to f_A tightens to $\mu_{B \rightarrow f_A} = \mu_{f_B \rightarrow B} = ([0.80, 0.10], [0.90, 0.20])$. Re-executing $f_A \rightarrow B$ with this informative B message and the still-clamped E produces a tighter interval. After message passing converges, the bounds on B follow from the extraction step applied to evidence-clamped messages:

$$\underline{P}(B=x \mid E=1) = \max_{f \in \{f_B, f_A\}} l_{f \rightarrow B}(x), \quad \overline{P}(B=x \mid E=1) = \min_{f \in \{f_B, f_A\}} u_{f \rightarrow B}(x).$$

f_B still contributes its prior bounds $[0.10, 0.20]$ on $P(B=1)$, while f_A now carries information about B conditioned on $E=1$. On this polytree the two are consistent and the conditional posterior recovers the prior:

$$P(B=1 \mid E=1) \in [0.10, 0.20],$$

reflecting the fact that B and E are marginally independent in the alarm network and the asymmetry of A 's CPT does not alter B 's marginal under evidence on E alone.

5.5 Comparison with Exact Variable Elimination

Query	Method	$\underline{P}$	$\overline{P}$
$P(B=1)$	Exact VE	0.1000	0.2000
$P(B=1)$	IBP	0.1000	0.2000
$P(A=1 \mid B=0, E=0)$	Exact VE	0.0000	0.0444
$P(A=1 \mid B=0, E=0)$	IBP	0.0000	0.0444
$P(B=1 \mid E=1)$	Exact VE	0.1000	0.2000
$P(B=1 \mid E=1)$	IBP	0.1000	0.2000

For root nodes like B , IBP recovers the exact bounds because the local factor alone determines the marginal. For conditional queries with evidence, the bounds may differ on loopy graphs due to the approximation inherent in message passing.

6 Multi-Valued Variable Support

A key advantage of IBP over L2U and GL2U is native support for multi-valued variables:

- **L2U**: Restricted to binary variables. Messages are scalar intervals.
- **GL2U**: Handles multi-valued variables by binarization (replacing a k -valued variable with $\lceil \log_2 k \rceil$ binary variables in a chain). This introduces auxiliary structure and can be lossy.
- **IBP**: Messages are interval vectors $(\mathbf{l}, \mathbf{u}) \in \mathbb{R}^k \times \mathbb{R}^k$. No binarization needed. Each state is bounded independently.

For example, the compound variable CD in the alarm network has 4 states. IBP directly propagates 4-dimensional interval vectors for CD , while GL2U would require binarizing CD into two binary variables.

7 Convergence Properties

- Messages are updated by taking min/max over all vertex and corner-point combinations, so the algorithm explores the full range of the credal set.
- On polytree-shaped networks with binary variables, IBP reduces to the 2U algorithm and converges to exact bounds.
- On loopy graphs, convergence to a fixed point is empirically observed but not formally guaranteed. The fixed point may not correspond to the exact bounds (similar to loopy belief propagation for Bayesian networks).

8 Complexity

Each iteration of IBP takes $O(N_f \cdot V_{\max} \cdot 2^{S_{\max}} \cdot k^{S_{\max}})$ time, where N_f is the number of factors, $V_{\max}$ is the maximum number of vertex combinations per factor, $S_{\max}$ is the maximum scope size, and k is the maximum domain size. The number of iterations until convergence depends on the network structure and is typically 10–100 for practical networks.

9 References

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